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Supply Chain Disruptions: Evidence from the Great East Japan Earthquake

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	The 2011 earthquake as a localized shock	4
3.2	Firm-level data and sample construction	4
3.3	Supply-chain distance	4
3.4	Outcome and controls	4
4	Empirical specification and identification	5
4.1	Baseline difference-in-differences network specification	5
4.2	Identification logic	5
5	Structural model and aggregation	5
5.1	Production-network mechanism	5
5.2	Reduced-form-to-structural mapping	6
5.3	Aggregate counterfactual	6
6	Tables and figures	6
6.1	Figure I: Geographical Distribution of Losses and Damages in Northeast Japan; Figure II: Growth Rate of Index of Industrial Production	6
6.2	Table I: Firm Size Distribution; Figure III: Industrial Composition and Geographical Distribution	7
6.3	Figure IV: Headquarters Locations of Firms in the Disaster Area; Table II: Pre-Earthquake Characteristics	8
6.4	Figure V: Propagation of the Shock over the Production Network; Figure VII: Robustness Checks	9
6.5	Figure VI: Headquarters Locations of Firms in the Flooded Region; Figure VIII: The Frequency Frontier; Figure IX: Energy Production	10
6.6	Figure X: Upstream and Downstream Propagation; Table III: Estimation of Elasticities of Substitution	11
6.7	Table IV: Aggregate Impact of a Potential Earthquake in the Tokai Region; Figure XI: Geographic Distribution of Firms Exposed to the Tokai Earthquake	12
7	Short summary	12
8	Conclusion	13
8.1	Core conclusion	13
8.2	Economic intuition	13
8.3	Incremental contribution revisited	13

8.4	Why this matters	13
8.5	Limitations	13
8.6	Takeaway	13

1 Big picture

- **Question.** How do large firm-level shocks propagate through production networks, and how much aggregate output loss can be attributed to supply-chain propagation?
- **Setting.** The paper studies the 2011 Great East Japan Earthquake, tsunami, and nuclear accident as a localized physical shock to firms in northeast Japan.
- **Data advantage.** The paper uses Tokyo Shoko Research firm-level data that identify customers and suppliers for hundreds of thousands of Japanese firms, together with firm sales and locations.
- **Core empirical design.** Firms outside the disaster area are treated if they are directly or indirectly connected to disaster-area firms through supplier or customer links. The paper compares their post-earthquake sales growth to firms farther away in the network.
- **Main empirical result.** The shock propagated both downstream to customers and upstream to suppliers, with effects declining in network distance but remaining visible several links away.
- **Theory contribution.** A general equilibrium production-network model explains why negative shocks propagate upstream when intermediate and primary inputs are gross complements, and downstream when intermediate inputs are gross substitutes.
- **Quantitative result.** Combining estimated elasticities with the firm-level network implies that the disaster lowered Japan's real GDP growth by about 0.47 percentage points in the year after the earthquake.
- **Policy/counterfactual result.** Applying the model to a potential Tokai earthquake implies GDP growth losses between roughly 0.48 and 2.17 percentage points depending on the exceedance-probability area.
- **Takeaway.** Micro supply chains can turn localized physical damage into macroeconomic losses. The structure of the production network is a propagation technology.

2 Incremental contribution of the paper

- **Causal firm-level evidence on network propagation.** The paper uses a large natural disaster as an exogenous, spatially concentrated shock and follows its propagation through actual firm-to-firm production links.
- **Both upstream and downstream propagation.** Prior network work often emphasizes supplier disruptions to customers. This paper shows that shocks also propagate upstream to suppliers through reduced demand.
- **Indirect network effects.** The paper documents that not only direct customers and suppliers, but also firms several links away, experience lower sales growth.
- **Rich network data at national scale.** The TSR data allow the authors to map customers and suppliers for a large fraction of Japanese firms, making it possible to study propagation in the observed production network rather than in industry-level input-output tables alone.
- **Reduced-form plus structural synthesis.** The paper does not stop at event-study evidence. It builds and estimates a production-network model to interpret the signs of propagation and quantify aggregate GDP effects.
- **Elasticity interpretation.** The estimated model recovers two key elasticities: intermediate inputs are gross substitutes while primary and intermediate inputs are gross complements. These estimates rationalize the empirical propagation patterns.
- **Disaster-risk counterfactual.** The paper uses the model to evaluate future earthquake scenarios, showing how micro-network exposure can convert hazard maps into macroeconomic loss estimates.

Interpretation: The incremental contribution is the combination of quasi-experimental shock variation, micro production-network data, and a structural aggregation exercise. The paper measures not merely whether networks matter, but how far, in which directions, and how much for aggregate output.

3 Data and measurement

3.1 The 2011 earthquake as a localized shock

The Great East Japan Earthquake struck on March 11, 2011, followed by a tsunami and the Fukushima nuclear accident. The disaster caused severe physical damage concentrated in coastal municipalities in northeast Japan. The officially defined disaster area consists of 41 municipalities.

Interpretation: The crucial empirical feature is spatial concentration. The shock was large locally, but the disaster-stricken prefectures accounted for only a small share of national GDP. This makes observed nationwide effects difficult to explain without propagation.

3.2 Firm-level data and sample construction

The Tokyo Shoko Research data contain information on firm sales, employment, location, industry, and buyer-supplier links. The main panel is restricted to firms with sales data for 2010, 2011, and 2012 and baseline covariates in 2010, yielding 750,237 firms.

Interpretation: The sample restriction creates a balanced three-year panel centered on the disaster. The authors validate representativeness by comparing firm-size, industry, and geographic distributions with the Economic Census.

3.3 Supply-chain distance

A firm is downstream distance k from the disaster area if it is a customer, customer-of-customer, and so on, k links away from disaster-area firms. A firm is upstream distance k if it is a supplier, supplier-of-supplier, and so on, k links away. The empirical specification truncates the distance dummies at $k = 4$.

Interpretation: Network distance is the treatment intensity. A firm may be physically far from the disaster but economically close if it buys from or sells to disaster-area firms.

3.4 Outcome and controls

The main outcome is log annual sales. Controls include pre-earthquake age, number of employees, number of transaction partners, and distance to the disaster area. The regressions include firm fixed effects and prefecture-industry-year fixed effects.

Interpretation: Prefecture-industry-year fixed effects absorb local and sectoral shocks. The remaining identifying variation compares firms in the same broad local-industry-year cells that differ in network distance to disaster-area firms.

4 Empirical specification and identification

4.1 Baseline difference-in-differences network specification

The core reduced-form regression can be written as

$$y_{ipst} = \alpha_i + \gamma_{pst} + \sum_{k=1}^4 \sum_{\tau \in \{2010, 2012\}} \beta_{k,\tau}^{down} \text{Down}_i^{(k)} \mathbf{1}\{t = \tau\} \quad (1)$$

$$+ \sum_{k=1}^4 \sum_{\tau \in \{2010, 2012\}} \beta_{k,\tau}^{up} \text{Up}_i^{(k)} \mathbf{1}\{t = \tau\} + X_i' \delta_t + \varepsilon_{ipst}. \quad (2)$$

Here y_{ipst} is log sales of firm i in prefecture p , industry s , and year t . The omitted year is 2011, so the 2012 coefficients measure post-earthquake growth relative to the control group, while 2010 coefficients test pre-trends.

Interpretation: The post-earthquake coefficients should be negative if the disaster propagated through the supply chain. The pre-earthquake coefficients should be close to zero if treated and control firms were on parallel trends before the shock.

4.2 Identification logic

The paper compares firms outside the four disaster-stricken prefectures that differ in their supply-chain distance to disaster-area firms. The identifying assumption is that, conditional on fixed effects and controls, network distance to the disaster area is unrelated to other unobserved shocks to sales growth.

- **Spatial contamination concern.** Firms physically near the disaster may be affected by infrastructure, electricity, or local demand disruptions rather than supply-chain links. The authors exclude firms in the four disaster-stricken prefectures and control for geographic distance.
- **Industry/prefecture shocks.** Prefecture-industry-year fixed effects absorb shocks common to firms in the same local industry.
- **Pre-trend test.** The 2010 interaction coefficients provide placebo tests for whether connected firms were already on different trends.
- **Robustness.** The authors redefine the disaster area using the flooded region, restrict to single-plant firms, restrict to western Japan to isolate electricity effects, and use a placebo production network.

Interpretation: The design is a network difference-in-differences: treatment is not physical exposure, but economic exposure through pre-existing trading links.

5 Structural model and aggregation

5.1 Production-network mechanism

The model has firms connected through intermediate input relationships. Each firm uses primary factors and intermediate inputs. Two elasticities govern propagation:

- σ : elasticity between primary inputs and the intermediate input composite;
- ξ : elasticity of substitution among intermediate inputs.

The empirical estimates imply

$$\hat{\sigma} \approx 0.595 < 1, \quad \hat{\xi} \approx 1.183 > 1.$$

Interpretation: Primary and intermediate inputs are gross complements, while firm-level intermediate inputs are gross substitutes. This combination produces upstream and downstream propagation patterns consistent with the reduced-form evidence.

5.2 Reduced-form-to-structural mapping

The paper maps reduced-form propagation coefficients into elasticities using the observed production network and assumptions about capital destruction in the disaster area. The baseline assumes $\mu = 0.5$ for the intermediate-input share and a 32.5

Interpretation: The structural exercise turns event-study coefficients into primitives of a production network model. It is not purely calibration; it uses the observed shock and network to estimate substitution parameters.

5.3 Aggregate counterfactual

The estimated model implies an aggregate GDP growth loss of about 0.47 percentage points from the 2011 disaster. The paper then uses the same machinery to assess potential Tokai earthquake scenarios.

Interpretation: The key result is that micro-level network disruption explains a large share of the observed national slowdown, even though direct physical damage occurred in a relatively small region.

6 Tables and figures

6.1 Figure I: Geographical Distribution of Losses and Damages in Northeast Japan; Figure II: Growth Rate of Index of Industrial Production

Read: Figure I maps fatalities/missing persons and demolished structures, showing that the disaster's physical damage was concentrated along the coast. Figure II shows the sharp decline in industrial production in the disaster-stricken prefectures after March 2011 relative to Japan as a whole.

Economic message: These figures establish the shock. It is spatially localized but economically severe, and manufacturing activity fell sharply where the physical shock hit. The contrast between the small GDP share of the directly affected region and the national slowdown motivates the supply-chain propagation hypothesis.

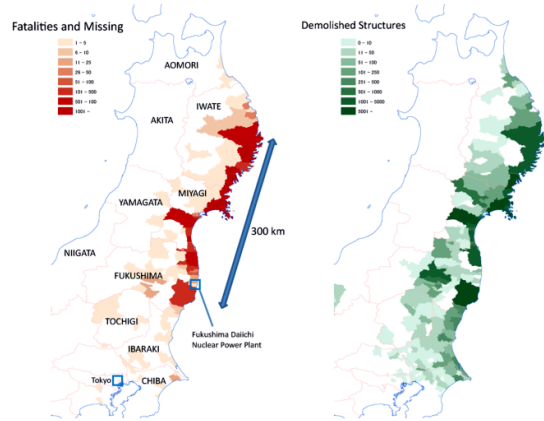


FIGURE I

Geographical Distribution of Losses and Damages in Northeast Japan

The figure depicts the distribution of casualties (left panel) and demolished structures (right panel) caused by the earthquake and its aftermaths at the municipality level. Color version available online.

(a) Figure I: physical losses and damages.

The first pair of visuals documents the localized physical shock and the real activity collapse in the disaster-stricken prefectures.

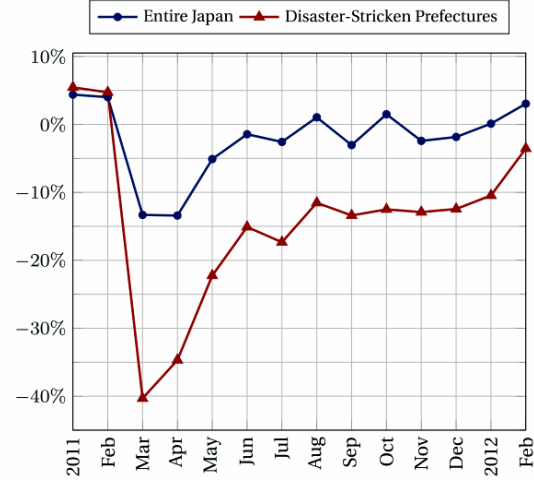


FIGURE II

Growth Rate of Index of Industrial Production

The figure plots the monthly year-on-year growth rate of the Index of Industrial

(b) Figure II: industrial production response.

6.2 Table I: Firm Size Distribution; Figure III: Industrial Composition and Geographical Distribution

Read: Table I compares firm-size bins in the TSR sample and Economic Census. Figure III compares industry and prefecture distributions. The main mismatch is that the TSR sample underrepresents very small firms.

Economic message: These visuals validate the empirical sample. The TSR data are not a random firm census, but the sample is broadly representative for firms with five or more employees. The credibility of network propagation evidence depends on using a large national firm sample. These visuals show that the analysis is not driven by a highly distorted sample composition.

TABLE I
FIRM SIZE DISTRIBUTION

	Number of employees									
	0-4	5-9	10-19	20-29	30-49	50-99	100-299	300-999	1,000-1,999	2,000+
TSR	45.9%	22.9%	14.4%	5.3%	4.5%	3.5%	2.5%	0.8%	0.1%	0.1%
Census	59.1%	17.1%	11.1%	4.2%	3.5%	2.6%	1.7%	0.5%	0.1%	0.1%

Notes. The table reports the fraction of firms with the number of employees in each of the respective bins. "TSR" refers to the 2010 TSR data set. "Census" refers to the 2009 Economic Census for Business Frame.

(a) Table I: firm-size distribution.

Sample validation visuals comparing TSR data to the Economic Census.

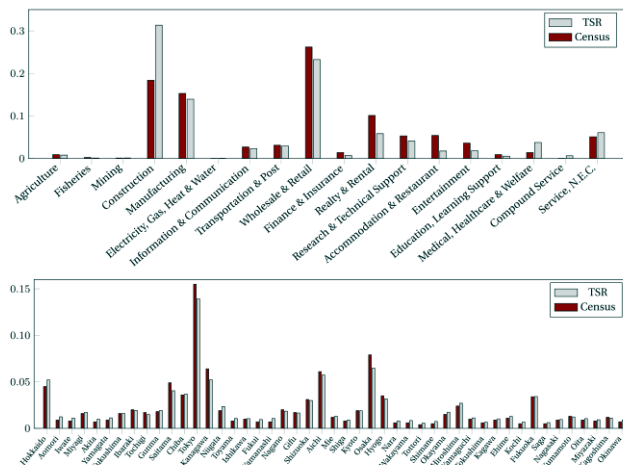


FIGURE III

Industrial Composition and Geographical Distribution

The top panel plots industrial composition (single-digit Japanese Standard Industry Codes) of firms. The bottom panel plots the share of firms in each of the 47 prefectures in Japan. "TSR" refers to the 2010 TSR data set. "Census" refers to the 2009 Economic Census for Business Frame. The vertical axis in each panel

(b) Figure III: industry and geography composition.

6.3 Figure IV: Headquarters Locations of Firms in the Disaster Area; Table II: Pre-Earthquake Characteristics

Read: Figure IV maps disaster-area firms. Table II compares pre-earthquake characteristics of firms inside and outside the disaster area.

Economic message: Disaster-area firms are similar to other Japanese firms in size, age, and network characteristics before the shock, while being spatially concentrated in physically affected areas. Table II is a balance table. It helps show that the shock is not simply hitting a special type of firm with unusual pre-existing characteristics.

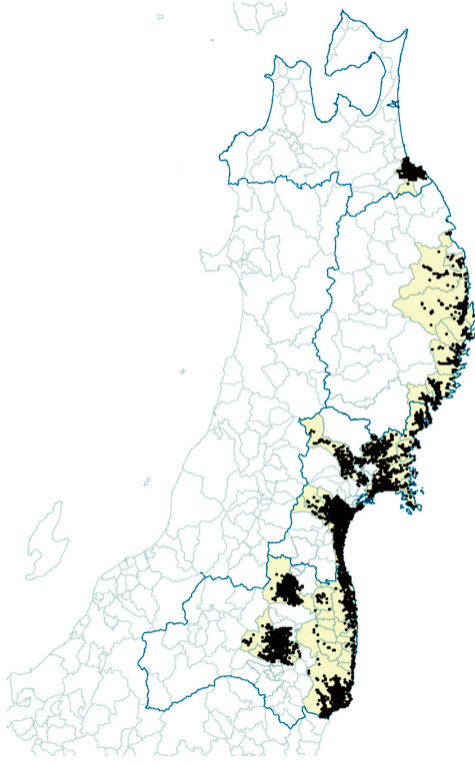


FIGURE IV

Headquarters Locations of Firms in the Disaster Area

(a) Figure IV: disaster-area firms.

The disaster-area definition and pre-shock firm comparison.

TABLE II
PRE-EARTHQUAKE CHARACTERISTICS OF FIRMS INSIDE AND OUTSIDE THE DISASTER AREA

	Disaster area				Rest of Japan
	Sorted by post-earthquake sales growth				
	All firms	Bottom tercile	Middle tercile	Top tercile	
Log sales	11.39 (1.67)	11.37 (1.64)	11.48 (1.69)	11.31 (1.68)	11.54 (1.77)
Log No. employees	1.83 (1.27)	1.74 (1.24)	1.88 (1.35)	1.88 (1.22)	1.84 (1.31)
Age	27.41 (15.48)	27.94 (15.45)	27.87 (15.80)	26.41 (15.12)	29.00 (16.54)
No. of suppliers	4.56 (14.68)	4.17 (12.54)	4.59 (12.18)	4.92 (18.17)	5.21 (34.99)
No. of customers	4.56 (30.37)	3.02 (6.36)	4.77 (37.70)	5.82 (36.12)	5.17 (36.14)
Customers' log sales	14.84 (2.43)	14.88 (2.43)	15.32 (2.44)	14.35 (2.32)	14.61 (2.53)
Suppliers' log sales	14.63 (2.39)	14.76 (2.45)	14.64 (2.45)	14.52 (2.30)	14.87 (2.56)

Notes. The table reports summary statistics of pre-earthquake characteristics of firms inside and outside the disaster area. Values are averages across firms in each category with standard deviations reported in parentheses. The first column reports means and standard deviations for all firms in the disaster area. The second, third, and fourth columns report means and standard deviations for disaster-area firms in, respectively, the bottom, middle, and top terciles of the distribution of the firms' post-earthquake sales growth rates. The last column reports means and standard deviations for firms outside of the disaster area.

(b) Table II: pre-earthquake characteristics.

6.4 Figure V: Propagation of the Shock over the Production Network; Figure VII: Robustness Checks

Read: Figure V reports the baseline network-distance coefficients. Post-earthquake estimates are negative for downstream and upstream firms, and effects decline with distance. Pre-earthquake estimates are statistically insignificant. Figure VII shows similar patterns under alternative definitions and sample restrictions.

Economic message: The core empirical result is propagation both downstream and upstream, not just to direct trading partners but also to indirect partners. The pre-trend diamonds in Figure V are the most important validity check. The blue post-earthquake coefficients are negative and significant; the red pre-earthquake coefficients are small and insignificant.

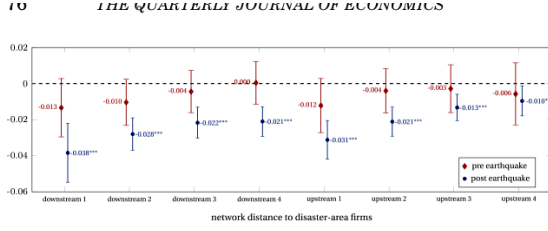


FIGURE V

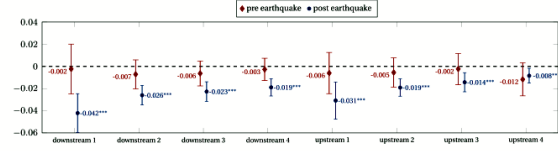
propagation of the Shock over the Production Network: Baseline Specification

This figure is based on a single panel regression, in which the dependent variable is the log of firms' annual sales. The plotted point estimates are the coefficients on network distance dummies interacted with annual dummy variables. The diamonds (in red) and circles (in blue) indicate the coefficients on network distance dummies interacted with 2010 and 2012 time dummies, respectively. The vertical bars indicate 95% confidence intervals based on two-way clustered standard errors at the level of prefecture and industry. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The regression includes firm and prefecture-year fixed effects and a set of control variables consisting of the log of the number of transaction partners, age, logarithm of the number of employees, and distance to the disaster area. All control variables are measured for the year 2010 and interacted with 2010 and 2012 time dummies.

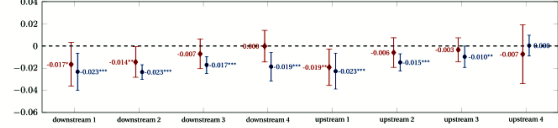
based on two-way clustered standard errors at the level of prefecture and industry.¹⁵ The coefficients on all downstream and upstream variables in the year following the earthquake

(a) Figure V: baseline propagation.

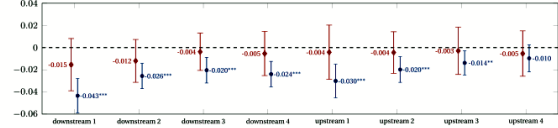
Baseline and robustness evidence for upstream/downstream propagation over the production network.



(A) Disaster area redefined as the region flooded after the tsunami



(B) Sample restricted to single-plant firms



(C) Sample restricted to firms west of the frequency frontier

FIGURE VII

Propagation of the Shock: Robustness Checks

(b) Figure VII: robustness checks.

6.5 Figure VI: Headquarters Locations of Firms in the Flooded Region; Figure VIII: The Frequency Frontier; Figure IX: Energy Production

Read: Figure VI redefines exposure based on flooded regions. Figure VIII uses Japan's 50/60 Hz frequency frontier to separate electricity-shock regions. Figure IX shows that energy production fell sharply in eastern regions but not western regions.

Economic message: These visuals address alternative channels. The authors show that their network results survive when exposure is defined more physically and when possible electricity-shortage confounding is handled. Pairing the frequency map and energy-production time series makes the electricity robustness design transparent: western Japan is a cleaner comparison because it was largely insulated from the eastern power shock.



FIGURE VI
Headquarters Locations of Firms in the Flooded Region
The map corresponds to the location of a firm in the TSR as the flooded region. The dark blue curve indicates the boundary of the flooded region (Fukushima, Iwate, and Miyagi) that encompass the disaster.



FIGURE VIII
The Frequency Frontier
The map depicts the partition of Japan's electric power system into the 50 Hz (lightly shaded area) and 60 Hz (unshaded area) regions. The darker areas near the boundary depict regions in which users of both frequencies exist.

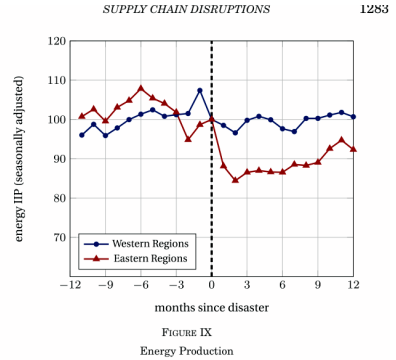


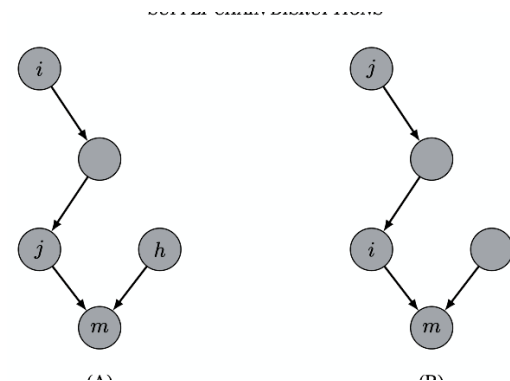
FIGURE IX
Energy Production
The figure plots the seasonally adjusted index of energy production for March 2010–February 2012. The index is normalized to 100 for February 2011. The horizontal axis refers to months since the disaster, with 0 corresponding to February 2011. "Eastern Regions" covers Hokkaido, Tohoku, and Kanto, all of which run on 50 Hz. "Western Regions" covers the rest of Japan, which runs on 60 Hz.

- (a) Figure VI: flooded region. (b) Figure VIII: frequency frontier. (c) Figure IX: energy production.
- Robustness visuals for alternative exposure definitions and electricity-shortage confounding.

6.6 Figure X: Upstream and Downstream Propagation; Table III: Estimation of Elasticities of Substitution

Read: Figure X illustrates stylized upstream and downstream propagation networks. Table III estimates the elasticities that determine propagation: $\sigma < 1$ and $\xi > 1$ across all specifications.

Economic message: The model explains why shocks move in both directions. Complementarity between primary and intermediate inputs helps upstream propagation; substitutability among intermediates helps downstream propagation. The table is the bridge from reduced-form evidence to macro quantification. The elasticities are statistically different from one and rule out a Cobb-Douglas benchmark.



(a) Figure X: upstream/downstream mechanisms.
Theory illustration and structural parameter estimates.

TABLE III
ESTIMATION OF ELASTICITIES OF SUBSTITUTION

	Baseline (1)	Capital destruction rate		Intermediate input share		Uniform- weighted network (6)
		Low (2)	High (3)	Low (4)	High (5)	
σ	0.595*** (0.062)	0.400*** (0.091)	0.694*** (0.047)	0.556*** (0.077)	0.639*** (0.051)	0.570*** (0.062)
ξ	1.183*** (0.034)	1.271*** (0.051)	1.138*** (0.026)	1.287*** (0.049)	1.111*** (0.024)	1.155*** (0.034)

Notes. The table reports estimates for the elasticities of substitution implied by regression specification (7), with robust standard errors reported in parentheses. *, **, and *** denote significance for the null hypothesis of the estimate being equal to 1 at the 10%, 5%, and 1% levels, respectively.

(b) Table III: elasticities of substitution.

6.7 Table IV: Aggregate Impact of a Potential Earthquake in the Tokai Region; Figure XI: Geographic Distribution of Firms Exposed to the Tokai Earthquake

Read: Table IV reports counterfactual GDP impacts for five nested Tokai earthquake exposure areas. Figure XI maps exposed firms under those scenarios.

Economic message: The framework turns hazard maps into macroeconomic loss estimates. A plausible Tokai event could lower GDP growth by 0.48 to 2.17 percentage points depending on the affected area. This counterfactual demonstrates the practical value of combining firm-level networks with geophysical risk maps.

SUPPLY CHAIN DISRUPTIONS 1299

TABLE IV
AGGREGATE IMPACT OF A POTENTIAL EARTHQUAKE IN THE TOKAI REGION

	Exceedance probability				
	50%	40%	30%	20%	10%
Number of disaster-area firms	27,812	40,258	59,187	74,548	95,640
GDP impact (percentage points)	-0.48	-0.88	-1.25	-1.51	-2.17

Notes. The table reports the number of disaster-area firms and the macroeconomic impact of a potential earthquake for five different scenarios. Each disaster area is defined by intersecting the area designated by the Central Disaster Management Council as the "Disaster Prevention Strengthening Area" with areas obtained from probabilistic seismic hazard maps. The exceedance probability is the probability that ground

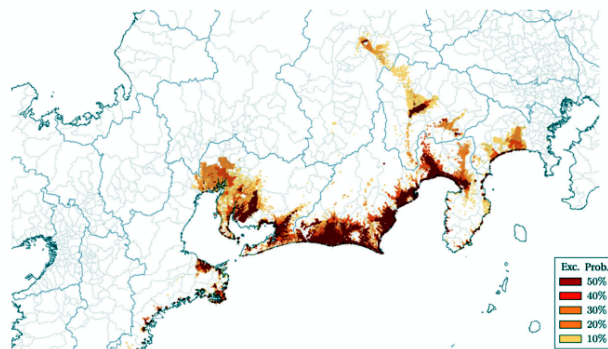


FIGURE XI
Geographic Distribution of Firms Exposed to the Tokai Earthquake

(a) Table IV: Tokai counterfactual impacts.

(b) Figure XI: firms exposed to Tokai earthquake.

Counterfactual macro impact of potential earthquakes in the Tokai region.

7 Short summary

- The paper studies the 2011 Great East Japan Earthquake as a localized shock to firms.
- It uses firm-level sales, location, and supply-chain data from Tokyo Shoko Research.
- Disaster-area firms are identified using 41 officially affected municipalities.
- Treatment is defined by upstream and downstream network distance to disaster-area firms.
- The core reduced-form evidence shows lower post-earthquake sales growth for firms connected to disaster-area firms.
- Effects occur for both suppliers and customers and decline with network distance.
- Pre-earthquake placebo coefficients are insignificant.
- Robustness checks use flooded regions, single-plant firms, the power-grid frequency frontier, and placebo networks.
- A structural model explains propagation through substitution and complementarity elasticities.
- Estimated elasticities imply intermediate inputs are gross substitutes while primary and intermediate inputs are gross complements.
- The disaster reduced Japan's real GDP growth by about 0.47 percentage points through network propagation.
- The model predicts sizable GDP losses from potential Tokai-region earthquakes.

8 Conclusion

8.1 Core conclusion

Carvalho, Nirei, Saito, and Tahbaz-Salehi show that production networks are quantitatively important channels for propagating real shocks. The Great East Japan Earthquake damaged firms in a localized region, but the economic effects travelled along buyer-supplier links to firms located far away from the disaster area.

8.2 Economic intuition

A firm hit by physical destruction cannot supply downstream customers and reduces demand for upstream suppliers. Those customers and suppliers then adjust production, passing the shock further through the network. Whether these effects propagate strongly depends on substitution elasticities. The estimated model implies that both upstream and downstream propagation are natural outcomes of the production structure.

8.3 Incremental contribution revisited

The paper’s main contribution is to combine a natural experiment, micro production-network data, and structural aggregation. The reduced-form estimates show propagation; the model explains the directions and magnitudes; the aggregation exercise quantifies macroeconomic losses.

8.4 Why this matters

Traditional regional disaster accounting may understate aggregate costs because it focuses on physical damage and local output. This paper shows that even firms with no physical exposure can suffer through network links. Disaster policy and macro risk assessment therefore require firm-level network information.

8.5 Limitations

The design relies on the exogeneity of network distance conditional on controls and fixed effects. Sales are observed annually, so high-frequency adjustment is not fully visible at the firm level. The structural model abstracts from many margins, including inventory buffers, contract renegotiation, and heterogeneous substitution across relationships.

8.6 Takeaway

Supply chains are not just conduits of goods; they are conduits of shocks. A localized disaster can become a macroeconomic event when it hits firms embedded in dense production networks.